Xin YANG

Department of Mechanical Engineering & Division of Mechatronic System Dynamics (LMSD), KU Leuven

CONTACT

- Personal page: https://xinyang-kul.github.io/;
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- Building G Room 106, Ghent and Aalst Campuses, Gebroeders De Smetstraat 1, 9000 Gent, Belgium;

EDUCATION

KU Leuven — Gent, Belgium
Ph.D. student (23'-), supervised by Prof. Dimitrios Chronopoulos — *Feb. 2023 - Present*

Tongji University — Shanghai, China
M.S. in Vehicle Engineering — *Sept. 2019 - Mar. 2022*

- **Cumulative GPA: 4.67/5.00**;
- **Scholarships and awards:** Graduate Entry Award, 2019; Outstanding Student Award (2021);
- **Course:** Vehicle dynamics@TJU, Modern Control Theory@TJU, Nonlinear Control Theory@TJU;

Chongqing University — Chongqing, China
B.S. in Vehicle Engineering — *Sept. 2015 - Jun. 2019*

- **Cumulative GPA: 3.68/4.00 (Ranking 5/144)**;
- **Scholarships and awards:** National Encouragement Scholarship, 2017; Outstanding Graduate of Chongqing University, 2019 ; Outstanding Student Award (top 3%), 2017 and 2018; Selected to Excellent Student Program (top 5%, on basis of outstanding academic performance);

JOURNALS

[1] ***Yang, X.****, Cantero-Chinchilla, S., Moradi, M., Komninos, P., Fang, C., Liao, Y.L., Kundu, P., Zarouchas, D., Chronopoulos, D. (2025). Damage imaging in structural health monitoring with fine-tuned conditional diffusion model. *Mechanical Systems and Signal Processing*, 236, 112996. doi:10.1016/j.ymssp.2025.112996. [paper]

[2] ***Yang, X.****, Fang, C., Kundu, P., Yang, J., Chronopoulos, D. (2024). A decision-level sensor fusion scheme integrating ultrasonic guided wave and vibration measurements for damage identification. *Mechanical Systems and Signal Processing*, 219, 111597. doi:10.1016/j.ymssp.2024.111597. [paper]

[3] ***Yang, X.****, Farrokhabadi, A., Rauf, A., Liu, Y., Talemi, R., Kundu, P., Chronopoulos, D. (2024). Transfer learning-based Gaussian process classification for lattice structure damage detection. *Measurement*, 238, 115387. doi:10.1016/j.measurement.2024.115387. [paper]

[4] Fang, C., ***Yang, X.***, Gryllias, K., Vandepitte, D., Liu, X., Zhang, L., Chronopoulos, D. (2025). Using removable sensors in structural health monitoring: A Bayesian methodology for attachment-to-attachment uncertainty quantification. *Mechanical Systems and Signal Processing*, 224, 111973. doi:10.1016/j.ymssp.2024.111973. [paper]

[5] Mardanshahi, A.*, Sreekumar, A., ***Yang, X.****, Barman, S.K., Chronopoulos, D. (2025). Sensing Techniques for Structural Health Monitoring: A State-of-the-Art Review on Performance Criteria and New-Generation Technologies. *Sensors*, 25(5), 1424. doi:10.3390/s25051424. [paper]

[6] Lu, H., Chinchilla, S., ***Yang, X.****, Gryllias, K., Chronopoulos, D. (2024). Deep learning uncertainty quantification for ultrasonic damage identification in composite structures. *Composite Structures*, 338, 118087. doi:10.1016/j.compstruct.2024.118087. [paper]

[7] Farrokhabadi, A., Lu, H., ***Yang, X.***, Rauf, A., Talemi, R., Chronopoulos, D. (2024). Energy Absorption Assessment of Recovered Shapes in 3D-Printed Star Hourglass Honeycombs: Experimental and Numerical Approaches. *Composite Structures*, 347, 118444. doi:10.1016/j.compstruct.2024.118444. [paper]

[8] Hu, X., ***Yang, X.***, Feng, F., Liu, K., Lin, X. (2021). A particle filter and long short-term memory fusion technique for lithium-ion battery remaining useful life prediction. *Journal of Dynamic Systems, Measurement, and Control*, 143 (6), 061001. doi:10.1115/1.4049234. [paper]

CONFERENCES

[1] ***Yang, X.***, Kundu, P., Chronopoulos, D. (2025). Fine-tuned Diffusion Model for High-resolution Sparse-Array Ultrasonic Guided Wave Imaging. *The 15th Prognostics and System Health Management Conference (PHM 2025)*. [paper]

[2] ***Yang, X.***, Chen, F. (2022). Deep Uncertainty Quantification of Prognostic Techniques for Proton Exchange Membrane Fuel Cell. *Vehicle Electrification and Powertrain Diversification Technology Forum Part II*. [paper]

[3] ***Yang, X.***, Chen, F., Jiao, J., Liu, S. (2021). Machine learning-based voltage degradation prediction with uncertainty quantifications for PEMFC. *2021 5th CAA International Conference on Vehicular Control and Intelligence (CVCI)*. [paper]

ACADEMIC ACTIVITIES

- Journal Reviewer:
  - Mechanical Systems and Signal Processing
  - IEEE Transactions on Industrial Informatics
  - IEEE Transactions on Instrumentation and Measurement
  - Expert Systems with Applications
  - Engineering Applications of Artificial Intelligence
  - Advanced Engineering Informatics
  - Measurement

WORKING EXPERIENCE

NIO Inc. Shanghai, China
Intern @ Vehicle Chasis Control Group *Feb. 2022 - Jun.2022*

- Lane centering control, Hardware in-the-loop testing, Ethernet/CAN communication protocol

Holomatic Inc. Shanghai, China
Intern @ Perception Group for Autonomous Driving *Sep. 2021 - Nov.2021*

- 3D object detection, Driving lane detection/segmentation

OTHER SKILLS

Coding: Python (Pytorch), MATLAB (Simulink), C++/C (LeetCode 300+), Linux(Ubuntu), Labview...

Languages: Mandarin, English (IELTS 7.0), Dutch (A2)

Expertise: Defect Detection, Deep/Transfer Learning, Data Analysis & Learning, Simulink System Modelling & Control...